ME 2900/3900/4900

Mission Control: Gesture-Guided Quadruped Robot Navigation

Creative Inquiry — Fall 2026

Course:	ME 2900/3900/4900
Credit hours:	1
Weekly meeting:	Fridays, 10:00 a.m.–12:00 noon, EIB 257
Modality:	In person

Project Advisor(s)

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Required Materials

Students must bring a laptop computer with:

- A functioning webcam and Wi-Fi for robot communications.
- The ability to install course software, including Python and VS Code.
- Windows 11 is acceptable, Ubuntu 20.04 is preferred.

All robot hardware, robot-camera access, shared software, and maze materials will be provided.

Course Description

Students work in teams addressing research and development problems under the supervision of a faculty advisor. Engineering principles and best practices are employed, with emphasis on teamwork, professionalism, experimental practice, documentation, and technical communication. Enrollment requires consent of the instructor.

This one-credit Creative Inquiry course introduces undergraduate students to human–robot teaming, computer vision, and safe high-level control of a quadruped robot. Students will also gain hands-on experience and basic understanding of Robot Operating System (ROS 2), the standard middleware used for communication, sensing, and control in modern robotic systems. Students will first develop and test hand-gesture recognition software using the webcams on their personal laptops. During robot operation, the laptop webcam will observe the operator's hand gestures, while a separate live video feed from the camera mounted on the Unitree Go1 quadruped robot will show the operator the environment in front of the robot. Each recognized gesture will be translated into a safe high-level motion command, such as move forward, move backward, turn, or stop. The final objective is to use this remote command interface to guide the Go1 safely through a modular maze without colliding with walls or obstacles.

Project Description

Mobile robots are often deployed into cluttered or hazardous environments where a human operator cannot maintain direct visual contact with the robot. Such scenarios arise in applications including search and rescue, disaster response, infrastructure inspection, precision agriculture, and defense operations, where direct human access may be unsafe or impractical. In these situations, the operator must understand the robot's surroundings through onboard video and issue commands from a remote mission-control station. Reliable human–robot collaboration is therefore essential for enabling robots to perform tasks safely and effectively in challenging environments. This project investigates an AI-enabled hand-gesture interface: the operator views the live video feed from the robot while computer-vision and AI models use the laptop webcam to recognize the operator's gestures. Motion-planning and control-design methods are then used to translate validated gestures into bounded robot velocity commands. A practical remote-command system must also address uncertain gesture recognition, video and communication delays, stale commands, obstacle proximity, and safe robot behavior when information is lost.

Students will develop a gesture-based command interface for the Unitree Go1 quadruped. Using their laptop webcams, teams will implement a starter vocabulary of stop, forward, backward, left, and right commands before integrating the interface with the robot's live video feed for remote maze navigation. Recognized gestures will be converted into bounded high-level velocity commands through the lab's existing control interface, with a safety layer that stops the robot when commands are stale or uncertain, communication or perception is lost, or an obstacle enters a defined stopping region.

Students will work in small teams to extend the starter system. Possible extensions include designing additional gestures, mapping gestures to instructor-approved body poses or gaits, and composing time-limited motion primitives. Each extension must remain interruptible by the stop command and must operate through the approved command and safety interfaces.

The final system will be evaluated in a student-built modular maze constructed from lightweight cardboard walls. The operator will remain at a separate command station and navigate using the robot's live video feed rather than direct visual observation of the maze. The required course activity will use a flat maze. Stair climbing and navigation through a debris field may be considered as optional advanced modules only after separate instructor review and validation.

Expected Student Activities

Students will work primarily in teams of two. Students will be expected to:

- Attend and participate in weekly project meetings.
- Learn the necessary introductory Python, computer vision, Git version control and robot communication concepts.
- Develop gesture-recognition software using a laptop webcam.
- Collect or use the provided image and video data to test gesture performance.
- Integrate laptop-based gesture recognition with the live video feed from the robot.
- Map validated gestures to approved, bounded robot actions.

- Implement and test command timeouts, confidence checks, stopping behavior, and other safety logic.
- Design and construct a modular maze within instructor-provided dimensional and safety constraints.
- Participate in integration tests and final maze-navigation trials.
- Maintain an individual research and design journal documenting decisions, results, problems, and contributions.
- Communicate progress through brief weekly updates, a final presentation, and a written report.

Students will begin with laptop-only development and recorded or live camera data. Access to live robot testing will be granted after the required software and safety checks have been completed.

Learning Outcomes

Upon successful completion of the course, students will be able to:

1. Describe the basic architecture of a robotic system, including sensing, computation, communication, control, and actuation.
2. Explain fundamental concepts in human–robot teaming, including how a human operator and a robot share information, decisions, and control responsibilities.
3. Apply basic robot-motion and control concepts, including body-frame velocity, turning rate, gait selection, and high-level command generation.
4. Use and evaluate camera-based AI models for hand detection and gesture recognition.
5. Apply introductory motion-planning and control-design concepts to translate recognized operator intent into bounded, safe robot velocity commands.
6. Integrate robot sensing, human commands, control logic, and robot hardware into a functional mechatronic system.
7. Incorporate safety constraints and fail-safe behavior into robot operation under uncertain perception, communication delays, and nearby obstacles.
8. Design and conduct repeatable engineering tests, analyze recorded data, and use evidence to evaluate performance and justify design decisions.
9. Contribute effectively to a collaborative robotics project using appropriate documentation, version-control, and project-management practices.
10. Communicate the motivation, design, results, limitations, and future directions of a robotics project to technical and nontechnical audiences.
11. Prepare and present project results through a research poster and live robot demonstration for audiences beyond the classroom.

Students will have the opportunity to present their work at Clemson University's annual Focus on Creative Inquiry (FoCI) poster forum. With advisor approval, projects that demonstrate sufficient technical maturity and research contribution may also be developed for submission to appropriate IEEE conferences or robotics competitions.

Weekly Activities Outline

The following 12-week activity outline represents the anticipated project progression. It intentionally leaves flexibility for university holidays, student preparation, make-up work, additional testing, and changes in project progress. The final two activity weeks are reserved for supervised robot demonstrations and maze navigation.

Week	Topic and activity
1	Project introduction, human–robot teaming objective, lab orientation, and robot safety.
2	Development environment, Python fundamentals, version control, and team workflow.
3	Image acquisition and visualization using laptop webcams.
4	Hand detection, landmarks, and the starter gesture vocabulary.
5	Gesture classification, confidence thresholds, temporal consistency, and performance testing.
6	Mapping gestures to abstract robot actions and testing commands in mock mode.
7	Robot camera feed, remote operator display, and high-level robot velocity commands.
8	Command timeouts, default-stop behavior, and obstacle-proximity stopping.
9	Integration of gesture recognition, remote video, motion commands, and safety supervision.
10	Modular maze design, supervised robot integration, and demonstration-readiness review.
11	Robot Demonstration I: system checkout, practice maze navigation, and corrective testing.
12	Robot Demonstration II: final team maze navigation, presentation, and project review.

Course Grading Policy

Students are expected to contribute approximately three hours per week for the one credit hour of enrollment. There are no examinations. Grades will reflect consistent individual engagement, technical progress, teamwork, documentation, safe lab practice, and communication.

Team results will contribute to the course grade, but each student must document and explain an identifiable individual contribution. A technically sound, well-documented, and safely tested system may receive full credit regardless of final robot demonstration.

The grade components are:

Component	Weight
Weekly participation, preparation, and progress updates	30%
Individual research and design journal	20%
Technical milestones, maze-navigation demonstration	40%
Final presentation, documented individual contribution, and written report	10%

Grading

Letter grade	Percentage
A	90–100%
B	80–89%
C	70–79%
D	60–69%
F	Below 60%

Deliverables

Students will complete the following deliverables:

1. **Weekly progress update:** A concise report of completed work, evidence, challenges, and next steps.
2. **Individual research and design journal:** A chronological record of technical work, decisions, test results, failures, and individual contributions.
3. **Gesture-recognition milestone:** A laptop-webcam demonstration of the starter gesture vocabulary with documented testing.
4. **Mock-command milestone:** Demonstration that gestures produce bounded abstract commands without operating the physical robot.
5. **Safety milestone:** Demonstration of default-stop behavior, command expiration, and obstacle-proximity stopping.
6. **Remote-interface milestone:** Demonstration of laptop gesture recognition operating alongside the live Go1 camera feed.
7. **Team maze design:** An instructor-approved modular maze that satisfies dimensional, visibility, and safety requirements.
8. **Final integration demonstration:** Supervised gesture-based navigation of the Go1 through an approved maze.
9. **Final presentation:** A team presentation describing motivation, design, implementation, testing, results, limitations, and future work.
10. **Final written report:** An individually attributable report or team report with clearly identified individual contributions, as specified by the instructor.

Lab and Robot Safety

Students must complete all lab and equipment training required by the project advisor before operating the robot or working in the robot test area. The following policies apply:

- Students may not operate any robots without instructor or authorized lab supervision.
- Laptop-webcam, mock-command, and safety milestones must be completed before live robot testing.

- Students may use only instructor-approved high-level robot actions and parameter limits.
- Direct low-level motor commands and arbitrary velocity values are prohibited.
- Stop is the default robot behavior, and every motion command must expire unless safely refreshed.
- A designated safety operator must be present during every live robot test with access to the approved emergency-stop method.
- People not assigned to the test must remain outside the marked robot workspace.
- Maze layouts and optional terrain modules must receive instructor approval before robot operation.
- Loose, sharp, unstable, rolling, or otherwise hazardous debris is prohibited.
- Damage, unexpected motion, loss of control, or unsafe conditions must be reported immediately.

Failure to follow lab or robot-safety procedures may result in suspension of hardware access and may affect the course grade.

Course Absence/Attendance Policy

Attendance and active participation in weekly meetings and scheduled lab sessions are required because the project depends on coordinated team development and supervised access to shared hardware. Students should notify the project advisor as early as possible regarding an anticipated absence and should use the Clemson Notification of Absence process when applicable.

An excused absence does not remove responsibility for communicating with the team, reviewing missed material, and completing an agreed-upon contribution. Two unexcused absences will result in a reduction of one letter grade in the student's final course grade.

AI Policy

Departmental placeholder: Insert the project-specific AI-use policy and current Fall 2026 Clemson guidance before submission.

Students will be expected to understand, test, document, and take responsibility for all code and technical content. Use of an AI tool will not substitute for the required individual explanation, validation, attribution, or demonstration of a student's contribution.

Title IX Statement

Departmental placeholder: Insert the current Clemson University Title IX statement before departmental submission.

Accessibility

Departmental placeholder: Insert the current Clemson University Student Accessibility Services statement before departmental submission.

Academic Integrity Statement

Departmental placeholder: Insert the current Clemson University Academic Integrity statement before departmental submission.

All software, reports, figures, datasets, and presentations must properly identify and cite material obtained from external sources. Collaboration within assigned teams is expected, but each student must accurately represent their individual contribution.

Emergency Preparedness Statement

Departmental placeholder: Insert the current Clemson University Emergency Preparedness statement before departmental submission.